

Sally's coffee Analysis

Ahmet Emin Yazgan

Aim of the project:

The aim of the project is to analyze Sally's Coffee's Sales and Customer Survey data to develop strategic recommendations to help increase profitability and revenue.

List of activities performed:

- Imported required libraries (Pandas, NumPy, matplotlib.pyplot, os)
- Warning messages are turned off with 'warnings.filterwarnings('ignore')'.
- User defined functions have been defined.
- first_looking(df): Analyzes the general information of the dataset (shape, column names, missing values, etc.).
- feature_looking(df, col): Shows the information of a single column, such as percentage of missing values, number of unique values, etc.
- missing_values(df): Finds missing values and calculates the percentage.
- duplicate_values(df): Checks for duplicate rows and removes them.
- get_state(zip_code): Determines the US state by ZIP code.
- The Survey dataset has been loaded into the Jupyter Notebook.
- Checked for duplicate data.
- The missing values were examined and some of them could be removed since the data we have is Survey data.
- Columns with 100% NaN values were dropped.
- Empty or incorrect data entries were corrected.
- Some categorical data were edited and transformed.
- The final version of the dataset was saved as Excel.
- Sales dataset was uploaded to Jupyter Notebook.
- In the Sales dataset, a State column was created according to Zip codes to help determine the relationships in Tableau.
- The updated Sales dataset was saved.
- Datasets were connected to Tableau and relationships were set up.
- The tasks specified by the Cafe owner were answered.
- Interactive dashboards were created using Sales and Survey data.

Key Insights

1- Coffee Consumption & Preferences (Survey Data)

Most respondents drink coffee at home.

Common brewing methods include pour over, French press, and espresso machines.

Many respondents have a coffee subscription, and the majority prefer light or medium roast.

Discounts seem to influence purchase behavior.

2- Roast Preference & Product Availability

Sally's own brand has fewer light roast compared to dark roast.

Competing brands offer significantly more light roast options.

This suggests an opportunity to expand Sally's light roast offerings to match consumer demand.

3- Sales Data (Geographic Analysis)

Sales are concentrated in New York, California, Washington, Rhode Island and Texas.

Certain ZIP codes drive significantly higher sales (e.g., 11354 and 10583 in New York).

Targeting these regions with localized marketing or discounts could further boost revenue.

Recommendations

Recommendations:

1. Reduce High Transportation Costs

-Optimize Distribution Centers: Establish regional fulfillment centers near high-demand regions (e.g. New York, Illinois).

-Negotiate Bulk Shipping Rates: Partner with logistics companies for volume-based discounts.

-Leverage Subscription Based Orders: Encourage more customers to subscribe to reduce single-order shipments.

-Estimated Annual Savings: A 5-10% reduction in transportation costs can save \$100,000 to \$500,000 annually, depending on order volume.

2. Increase Own Brand Coffee Sales

-Expand Light Roast Offerings: Increase the number of SKUs to compete with other brands.

-Offer Subscription Discounts: Offer discounts on Sally's brand for subscribers.

- Improve Brand and Packaging: Emphasize ethical sourcing and flavor notes to differentiate.
- Promote Special Blends: Create limited-time or seasonal flavors to attract more customers.

3. Strategic Actions:

- Exclusive Subscription Benefits: Offer only Sally's blends for subscribers.
- Provide tiered discounts (e.g. 10% off the first 3 months).
- Rebranding and Awareness Campaigns:
 - Highlight sustainable sourcing and unique flavor profiles.
 - Improve packaging design to visually differentiate Sally's coffee on the shelf.

4. Create More Effective Surveys

- There are many options left blank in the survey data and these should be avoided. The survey should be prepared in such a way that there are no blank questions.
- In the data, there are many surveys that are answered as I like coffee in one question and I don't like the taste of coffee in the other question or vice versa, if this can be prevented, more effective analyzes can be made.